

MBARI Waypoint Management System Test Plan

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Draft 1: 9/15/2000

Introduction

This document describes test procedures for the MBARI Waypoint Management System.

The database and the processes described here were developed to:

- Maintain a "master" list of waypoint names that can be used by both MBARI vessels for ROV operations.
- Provide mechanisms to update this list following shipboard activities.
- Allow for editing, insertion, and deletion of waypoints from the scientist's desktop.
- Provide a way to search and extract waypoints for import into other programs

The system is coordinated with MBARI's Expedition Database pre-cruise planning and post-cruise reporting system. The waypoints are for navigational use and may include midwater, benthic, and shore locations.

The system is intended to serve the institutional needs of MBARI and may become a core data stream. It is needed now to facilitate communication, prevent confusion, and avoid any misunderstandings of waypoint names or locations.

Antecedent documents include 1) MBARI Waypoint Management System Requirements Document, Draft5, 4/18/2000, 2) Design diagrams from Gerry, 3) Email discussion detailing design decisions. Please refer to the Requirements Document for process descriptions and a Use Scenario. The following sections detail the test procedures and the success criteria for each of the tests. (Draft1: the list is not complete.)

Shipboard use

Waypoint Query GUI

1. Query for list of waypoints based on geographic region, name, or owner:
Success: File suitable for input to Winfrog is created.

Winfrog

1. Operate Winfrog as usual, creating and editing user events that will become waypoints
Success: Normal operation of Winfrog is not affected, files are successfully saved and integrated with the waypoint management system.

UploadUserEvents GUI

1. Create user log file from Winfrog and upload this to the database:
Success: All relevant fields, including owner, are inserted or updated in the shipboard MSDE database. (Merge replication and Resolve Waypoints will handle conflicts.)

Shoreside use

Waypoint.asp

1. All web page interactions shall provide a response to the user in 3 seconds or less.
2. Search for waypoint based on owner:
Success: List returned to browser with options for edit, delete, and export
3. Edit fields of a waypoint from this returned list:
Success: Database table is updated, confirm by re-executing search query.
4. Delete waypoint from this returned list:
Success: Record removed from table

5. Add or edit giving a duplicate waypoint name:
Success: Message returned noting that waypoint exists and another name should be chosen.
6. Export list of waypoints as comma separated value file:
Success: Waypoints successfully imported into ArcView and Excel
7. Import list of Waypoints from a comma separated value file:
Success: Waypoints in file from users desktop successfully entered into database.

Resolve Waypoints

1. Create conflict with same WaypointName generated on shore, on the Western Flyer, and on the Point Lobos:
Success: Shipboard modified or created waypoint in entered into database. Email is sent to owner of waypoint that was replaced.

Expire

1. Set expiration date field to force an expire:
Success: Waypoint record is removed from the waypoint table and inserted into the histroy table. Email is sent to the owner.

Merge Replication

1. Both ships connected to network:
Success: Databases are replicated.
2. Point Lobos disconnected from network:
Success: User applications run normally

Special cases to test:

- a) We need to be sure that all computers which will be participating in the replication will are running SQL Server and MSDE as LAGOON\sqlxec. (And that LAGOON\sqlxec is an Administrator on each of these machines.)
- b) We need to verify that the database will perform as needed when the ship computers are not connected to the shore (lagoon domain).

2. Western Flyer disconnected from network:

Success: User applications run normally

2. Ships connect back to the network:

Success: The 3 databases are synchronized, conflicts are resolved.